

Rheumatoid Arthritis

SLE

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Rheumatoid arthritis (RA)

Chronic

System

Inflammatory disorder

Unknown aetiology

Primarily involves
synovial joints



Rheumatoid arthritis (RA)

Typically symmetrical

Destruction of joints –
erosion of cartilage and bone

Causing joint deformities

Progresses from the
periphery to more proximal
joints

Significant locomotor
disability within 10 to 20
years - who do not “fully”
respond to treatment.



Rheumatoid arthritis (RA)

INITIAL CLINICAL PRESENTATION

Polyarticular disease -

Onset

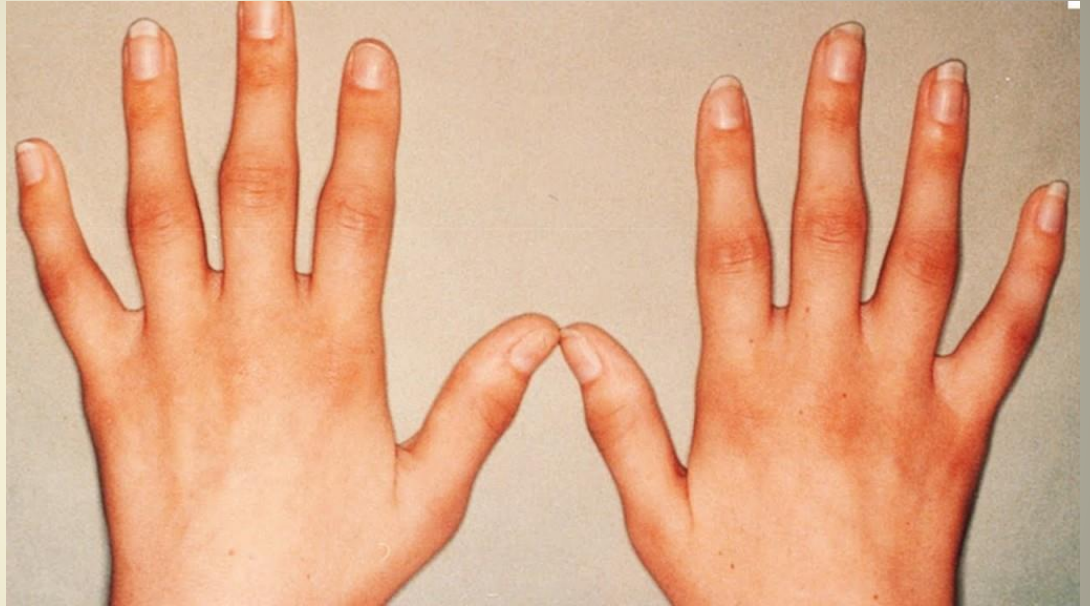
Gradual

Acute

Intermittent

“Migratory”

Monoarticular disease



Rheumatoid arthritis (RA)

Symptoms

Affect capacity to perform the activities of daily living – ADL:

Walking

Stairs

Dressing

Use of a toilet

Getting up from a chair

Opening jars

Doors

Typing

Type of job they have

Ability to do their job



Rheumatoid arthritis (RA)

Painful inflammation –
local tenderness from pressure applied
on the joint
pain on moving the joint

Swelling –
synovial hypertrophy
effusion.

Synovial thickening –
"boggy" feel to a swollen joint

Effusion –
Fluctuant

Heat and redness are not prominent
features

Although -involved joint is often
perceptibly warmer on examination



Rheumatoid arthritis (RA)

Synovitis of the wrist - carpal tunnel syndrome even in very early disease,

When swelling may not be particularly evident

The characteristic joint deformities -late manifestations of disease



Rheumatoid arthritis (RA)

Distribution of involved joints —
eventually affects - peripheral joints in almost all patients.

Involvement of axial and central joints is less common, occurring in 20 to 50 percent of patients

Interfacetal / atlantoaxial joints of the neck

Acromioclavicular

Sternoclavicular

Temporomandibular,

Cricoarytenoid

Shoulders

Hips

Lumbar spine facet joint synovitis may occur -rare.



Rheumatoid arthritis (RA)

Hands -

Early

Tender

Restricted range of movement

Reduced grip strength

Palmar erythema flexor tendons -
synovitis of the tendon sheaths
"tenosynovitis"

Nodules

Ulnar deviation

Swan neck

Boutonniere



Rheumatoid arthritis (RA)

Feet and ankles -

MTP joints - common in early disease

- Tenderness - resulting in the tendency to bear weight on the heels and hyperextend the toes.

- Erosive damage - lateral drift of the toes and plantar subluxation of the metatarsal heads resulting in "cock-up" deformities
 - bony lumps on the sole with associated callosities.

- Tarsus + tendon sheaths - pain on inversion or eversion - diffuse oedema and erythema over the dorsum of the foot.

- Heel pain -retrocalcaneal bursitis
tarsal tunnel syndrome-impingement of the posterior tibial nerve.
= paresthesiae of the toes

- Ankle -diffuse swelling around the tibiotalar -wrongly attributed to fluid retention



Rheumatoid arthritis (RA)

LABORATORY FINDINGS

- **Synovial fluid** - leukocyte - polymorphonuclear
- **Hematologic** – anaemia of chronic inflammation, thrombocytosis, mild leukocytosis.
- **Autoantibodies** –75 to 80 % - RF + Anti-CCP Ab
- **Acute phase response** – ESR - CRP



Rheumatoid arthritis (RA)

IMAGING -

Plain film radiography

MRI - identifying bone erosions -estimate the quantity of hypertrophic synovial tissue

Ultrasonography - degree of inflammation and the volume of inflamed tissue



Rheumatoid arthritis (RA)

CLINICAL COURSE — marked variation of clinical expression

- number of involved joints
- pattern of joint involvement
- fluctuations in disease activity
- ability to achieve remission
- rate of progression
- extent of structural damage.



Rheumatoid arthritis (RA)

Patterns of progression — Variation

- Most - fluctuation of disease activity over periods lasting weeks to months - increase or decrease in symptoms - pattern - recur throughout the course of the disease.

- Remission - small proportion of patients

Disease activity versus structural damage —

disease activity = underlying inflammatory response

destructive process = irreversible damage of the joint



Rheumatoid arthritis (RA)

- Disease activity - variation - endogenous rhythms of the disease - spontaneous exacerbations and quiescence, "flare"
- Structural damage is cumulative and irreversible - linked to inflammation - hence to disease activity, but – degeneration and repair.

As structural damage progresses - detection of variation in disease activity by clinical examination - increasingly difficult.

At these later stages, symptoms and signs - continuing rheumatoid disease or mechanical and degenerative change.



Rheumatoid arthritis (RA)

Therapies

Pharmacological

NSAIDS / COXibs

DMARDs

Biologics

Non – Pharmacological

Physiotherapy

Occupational Therapy

Complimentary



Systemic lupus erythematosus (SLE)

Chronic inflammatory disease

Unknown cause

Affect virtually every organ

Immunologic abnormalities -
antinuclear antibodies

Women are affected more
frequently than men.



Systemic lupus erythematosus (SLE)

SLE -protean clinical manifestations

- vary dramatically from patient to patient.

The most common pattern is a mixture of constitutional complaints –

Skin

Musculoskeletal

Mild hematologic

serologic involvement.

Some - predominately

Hematologic

Renal

Central nervous system

The pattern that dominates during the first few years of illness tends to prevail throughout the disease course.

clinical course - highly variable - periods of remissions and of chronic or acute relapses.



Systemic lupus erythematosus (SLE)

Constitutional symptoms

Fatigue

Fever

Myalgia

Weight change



Systemic lupus erythematosus (SLE)

Arthritis and arthralgias

Arthralgia 90 %

- earliest manifestations

Arthritis – inflammation –

- 65 -70 %

- migratory

- polyarticular,

- symmetrical

Moderately painful

Usually does not cause erosion

Rarely deforming.

Deforming erosive arthritis - similar -
rheumatoid arthritis.



Systemic lupus erythematosus (SLE)

Skin and mucous membrane involvement

Most patients - some point

- facial eruption - “the butterfly rash”

- discoid lesions - inflammatory -
tendency to scar

Photosensitivity

Oral and/or nasal ulcers - painless,

Non-scarring alopecia



Systemic lupus erythematosus (SLE)

Raynaud phenomenon

vasospastic process

50 %

Warmer to Cooler environment



**Lacking in blood flow, skin discolors
& takes a blue or white tinge**

Systemic lupus erythematosus (SLE)

Vasculitis

11 -36 %

Clinical spectrum – broad - potential for inflammatory involvement of vessels of all sizes

Small vessel - most common - cutaneous lesions

Palpable purpura

Petechiae

Papulonodular lesions

Livedo reticularis

Panniculitis

Splinter hemorrhages

Superficial ulcerations

Medium and large vessel

Mesenteric

Hepatic

Pancreatic

Coronary

Pulmonary

Retinal

Peripheral/ CNS



Systemic lupus erythematosus (SLE)

Thromboembolic disease

Venous / Arterial

Arterial thrombotic event (ATE) -
11 %

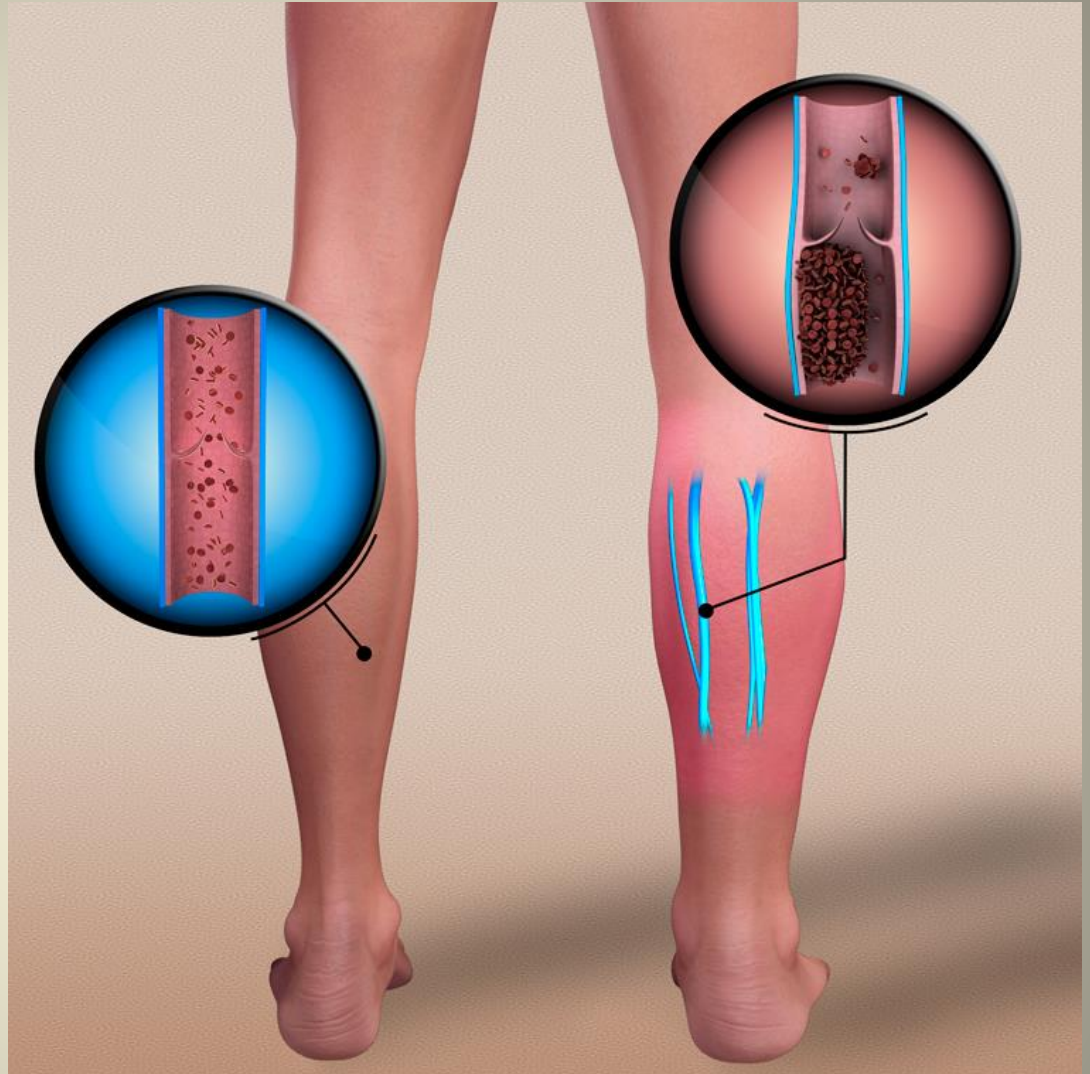
Venous thrombotic event (VTE) –
5%

Estimated 10-year risks

- 10 % for VTE,

- 26 % for ATE,

- 33 % for any thrombotic event



Systemic lupus erythematosus (SLE)

Renal involvement

50 % - significant cause of morbidity and mortality

Periodic screening

- Urinalyses
- Quantitation of proteinuria
- GFR

Important component of the ongoing management

Several forms of glomerulonephritis -
Biopsy

Clinical presentation - highly variable –

- Asymptomatic
- Hematuria
- Proteinuria
- Nephrotic syndrome
- Rapidly progressive glomerulonephritis
- Renal transplant / Dialysis
- Hypertension.



Systemic lupus erythematosus (SLE)

Gastrointestinal involvement

40 %

Majority - adverse medication reactions

viral / bacterial infections

Esophagitis

Intestinal pseudo-obstruction

Protein-losing enteropathy

Lupus hepatitis

Acute pancreatitis

Mesenteric vasculitis / ischemia

Peritonitis



Systemic lupus erythematosus (SLE)

Pulmonary involvement

Pleuritis

Pneumonitis

Interstitial lung disease

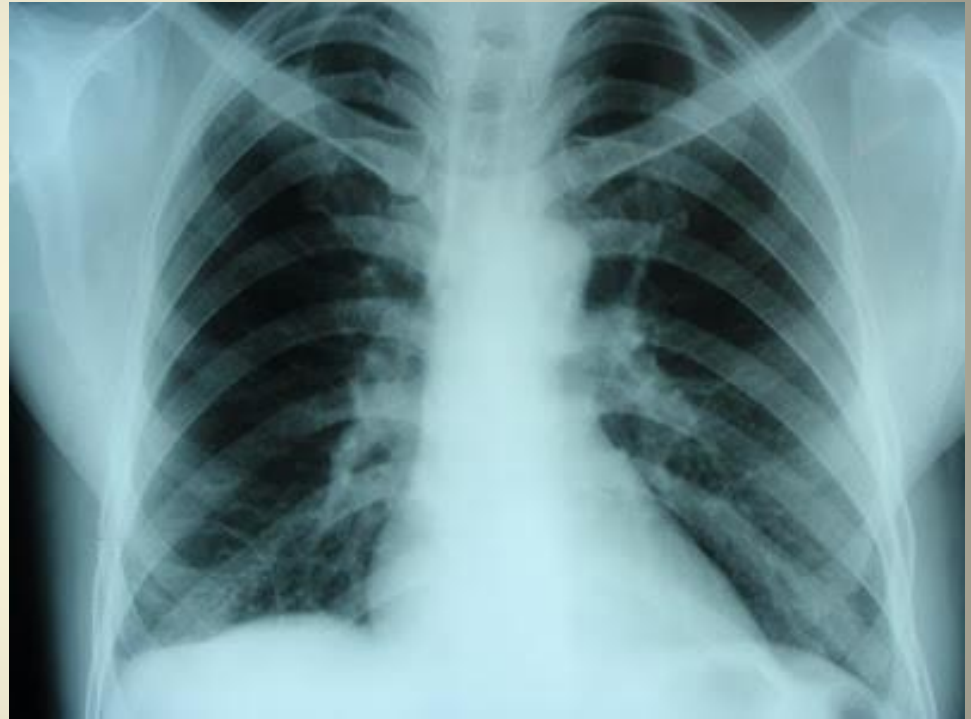
Pulmonary hypertension

Shrinking lung syndrome,

Alveolar haemorrhage.

Respiratory symptoms -
distinguished from infection -
particularly - immunosuppressive
therapy.

Thromboembolic - increased -
antiphospholipid antibodies /
lupus anticoagulant.



Systemic lupus erythematosus (SLE)

Neuropsychiatric involvement

Broad range of neurologic and psychiatric manifestations

Cognitive dysfunction

Organic brain syndromes

Delirium

Psychosis

Seizures

headache

&/or peripheral neuropathies

Cranial neuropathies

Myelitis

Meningitis

Psychosis – SLE or Steroid

Depression

Anxiety

Mania

Thromboembolic events - stroke



Systemic lupus erythematosus (SLE)

Ophthalmologic involvement

Common

Keratoconjunctivitis sicca

Less Common

Retinal vasculopathy

Rare

Choroidopathy

Episcleritis

Scleritis

Anterior uveitis -iritis

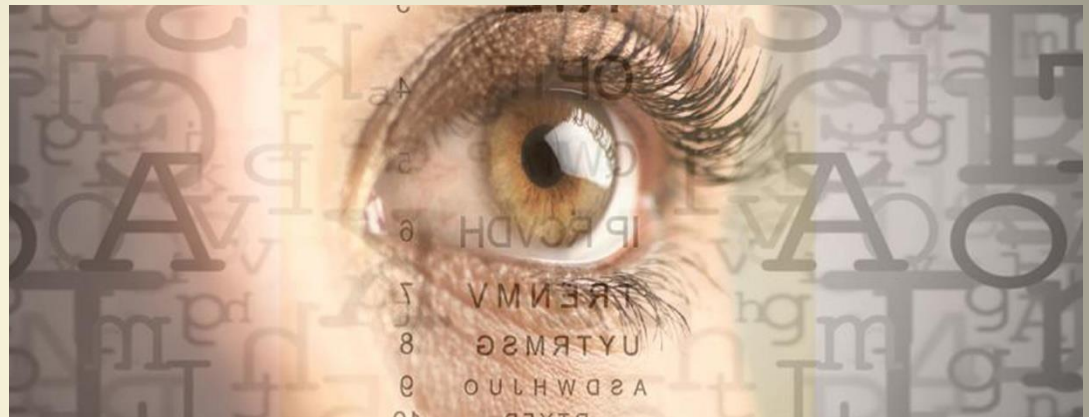
Orbital tissues

Pain

proptosis

Lid swelling

Diplopia



Systemic lupus erythematosus (SLE)

Hematologic abnormalities

Anaemia of chronic disease - most common

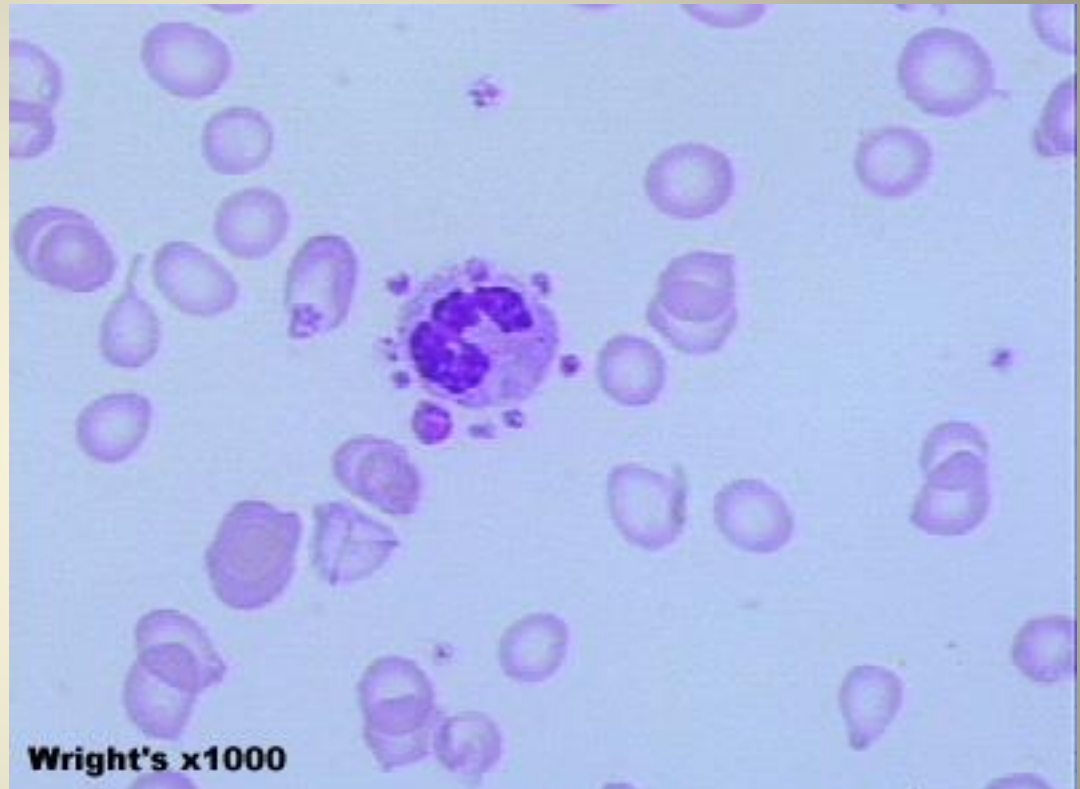
Autoimmune hemolytic anemia - relatively rare

- Leukopenia - 50 %

lymphopenia and/or secondary neutropenia

- correlates with clinically active disease

- Mild thrombocytopenia - common



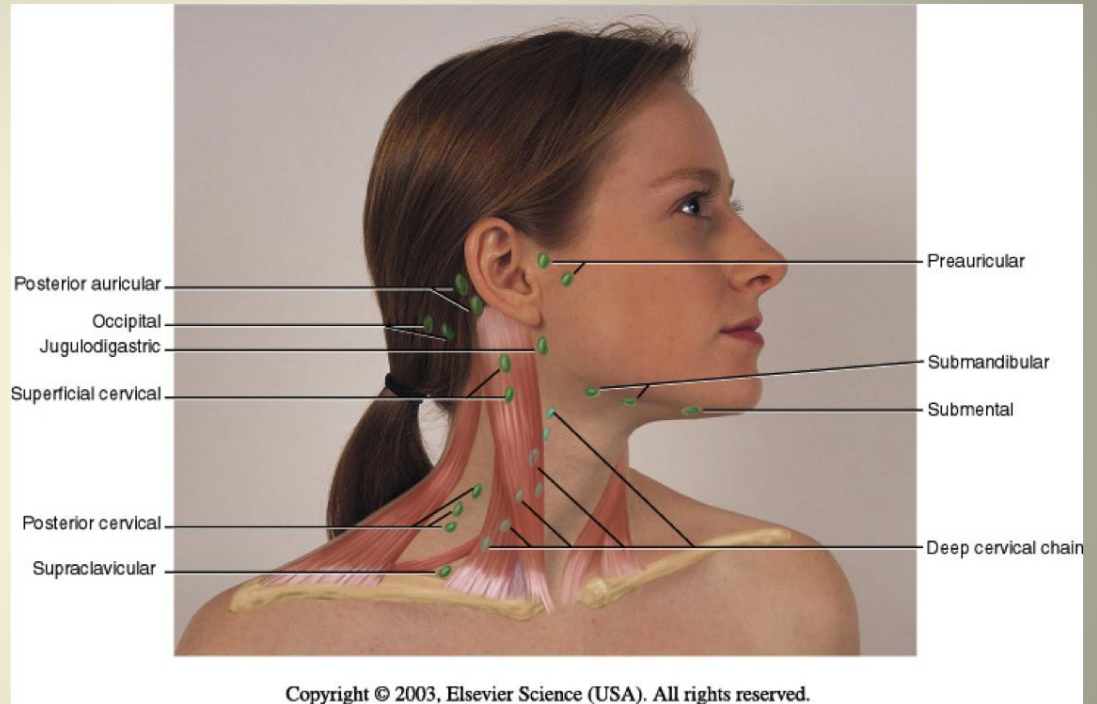
Systemic lupus erythematosus (SLE)

Lymphadenopathy and splenomegaly

Cervical

Axillary

Inguinal



Systemic lupus erythematosus (SLE)

Cardiac disease

Pericardium - 25 %

Myocardium - uncommon

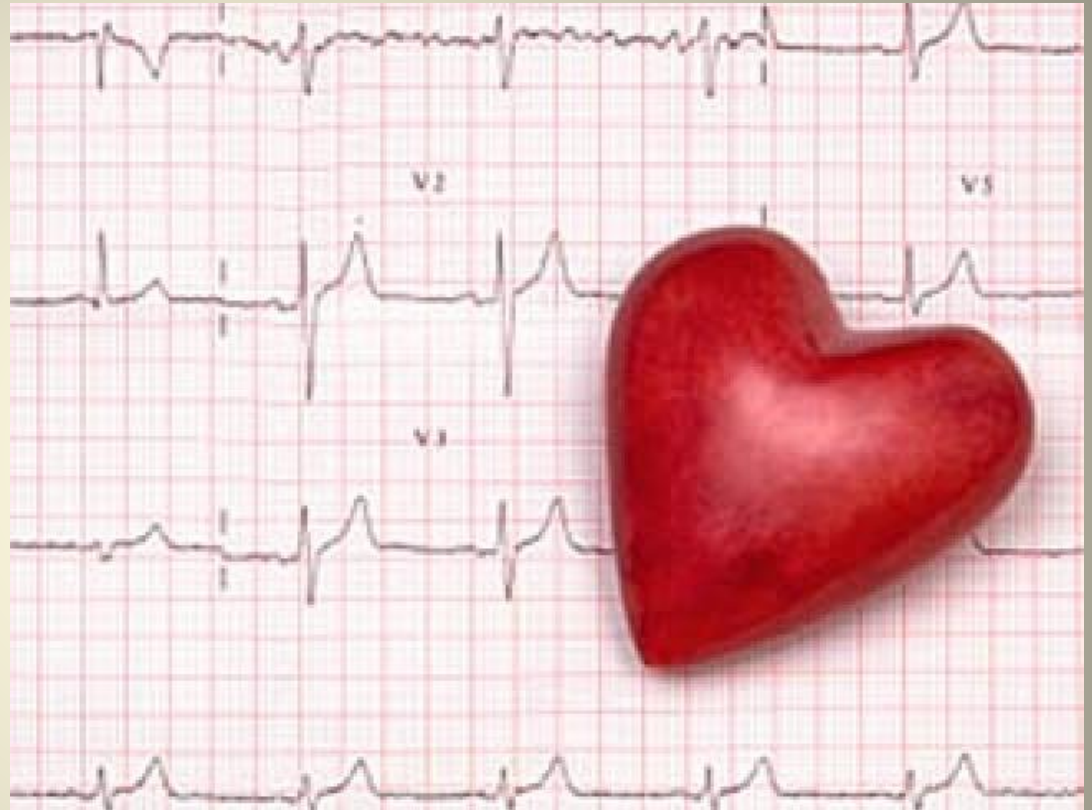
Valves - Libman-Sacks
endocarditis

Conduction system

Coronary arteries.

Note:

Neonatal lupus - babies of women
with SLE - anti-Ro/SSA and anti-
La/SSB = cause heart block of
varying degrees - congenital heart
block





Reflections and Questions ...